An introduction to the package MetabolSSMF

Contents

Introduction

This vignette presents the **MetabolSSMF** package, which implements a framework for simplex-structured matrix factorisation (SSMF) algorithm in R. This package provides a standard algorithm, which allows the users simply to apply the SSMF to perform soft clustering on the dataset that they are interested in. The usage of this package will be illustrated through an application to the simulated metabolomic dataset. This vignette also includes the usage of:

- **gap statistic**, a guidance to select the number of clusters;
- **bootstrap resampling** approach for estimating confidence intervals;
- **soft adjusted Rand index** for comparing known and estimated partitions;
- **Shannon diversity index** for evaluating the average uncertainty in soft clustering partition.

This document gives a quick tour of **MetabolSSMF** functionalities. It was written in R Markdown, using the knitr package for production. See help(package="MetabolSSMF") for further details.

Installation: The latest version of the package can be installed from **Github**:

```
# Install the package
devtools::install_github("WenxuanLiu1996/MetabolSSMF")

# Load the package
library(MetabolSSMF)
```

Data

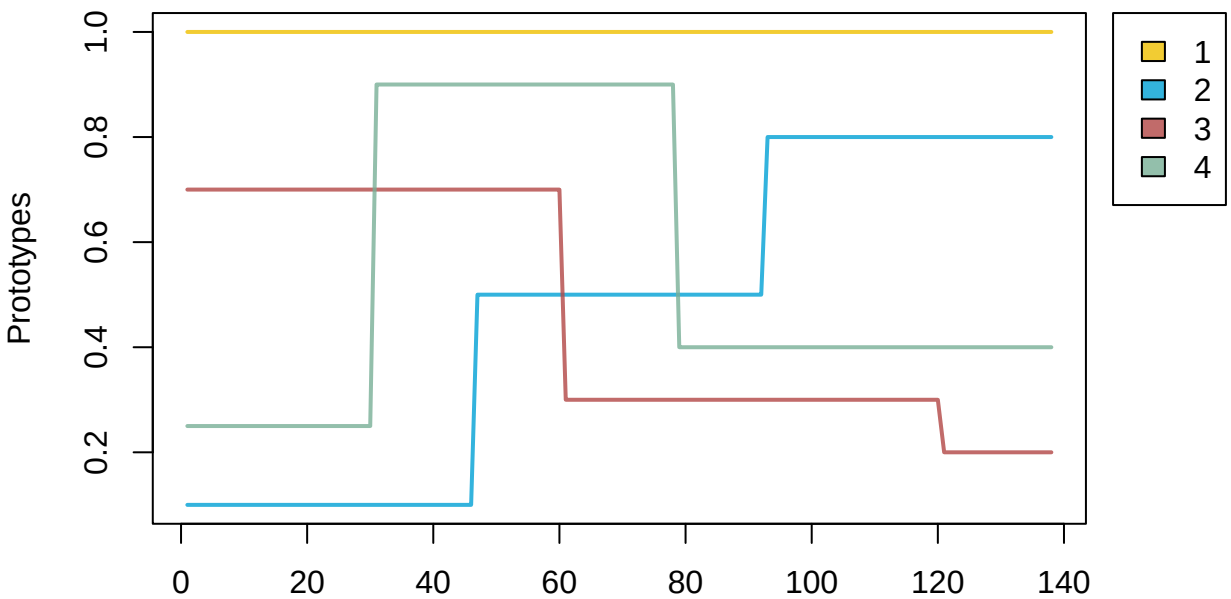
The simulated dataset is created by the product of a soft membership matrix H and a prototype matrix W. User can use a Dirichlet distribution to generate the values for the soft membership matrix H with simplex structure. W can be defined using designed pattern. The package provides the simulated dataset X, simulated prototype matrix W and simulated membership matrix H. They can be loaded through the code `data(SimulatedDataset)`, `data(SimulatedPrototypes)` and `data(SimulatedMemberships)`, respectively.

```
# Simulated data set X
X <- as.matrix(SimulatedDataset)

# Simulated W matrix (prototype matrix)
W <- as.matrix(SimulatedPrototypes)

# Simulated H matrix (membership matrix)
H <- as.matrix(SimulatedMemberships)
```

Visualise the simulated data

Visualise the values of simulated prototype matrix W

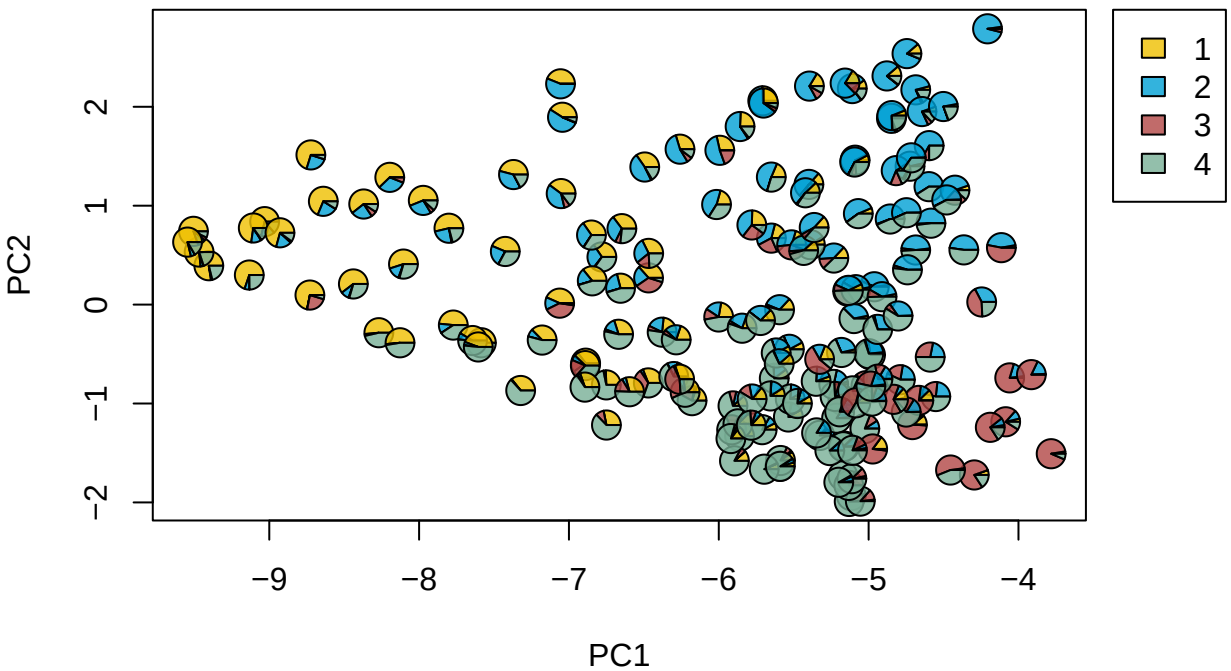
```
# Package for colour palettes
library(ggsci)
# Define 4 different colors
color <- c(pal_jco("default", alpha = 0.8)(7)[2], pal_jama("default",
    alpha = 0.8)(7)[c(3:5)])

par(mar = c(5.1, 4.1, 4.1, 4.1), xpd = TRUE)
matplot(t(W), type = "l", lty = 1, ylab = "Prototypes", col = color,
    lwd = 2)
legend("topright", inset = c(-0.15, 0), legend = 1:4, fill = color[1:4])
```



Visualise the values of simulated soft membership matrix H in each observation

The scatter pies plot of 2-dimensional projected clusters. It demonstrates the simulated data with 4 soft clusters, the area of colour in the pies depict the values of soft membership in the observations

```
# Package for pies plot
library(caroline)
pca_X <- prcomp(X, center = F, scale. = F)
par(mar = c(5.1, 4.1, 4.1, 4.1), xpd = TRUE)
pies(lapply(apply(H, 1, list), unlist), color.table = nv(color[1:4],
    paste0("Pty", 1:4)), x0 = pca_X$x[, 1], y0 = pca_X$x[, 2],
    radii = 2, ylab = "PC2", xlab = "PC1")
legend("topright", inset = c(-0.15, 0), legend = 1:4, fill = color[1:4])
```



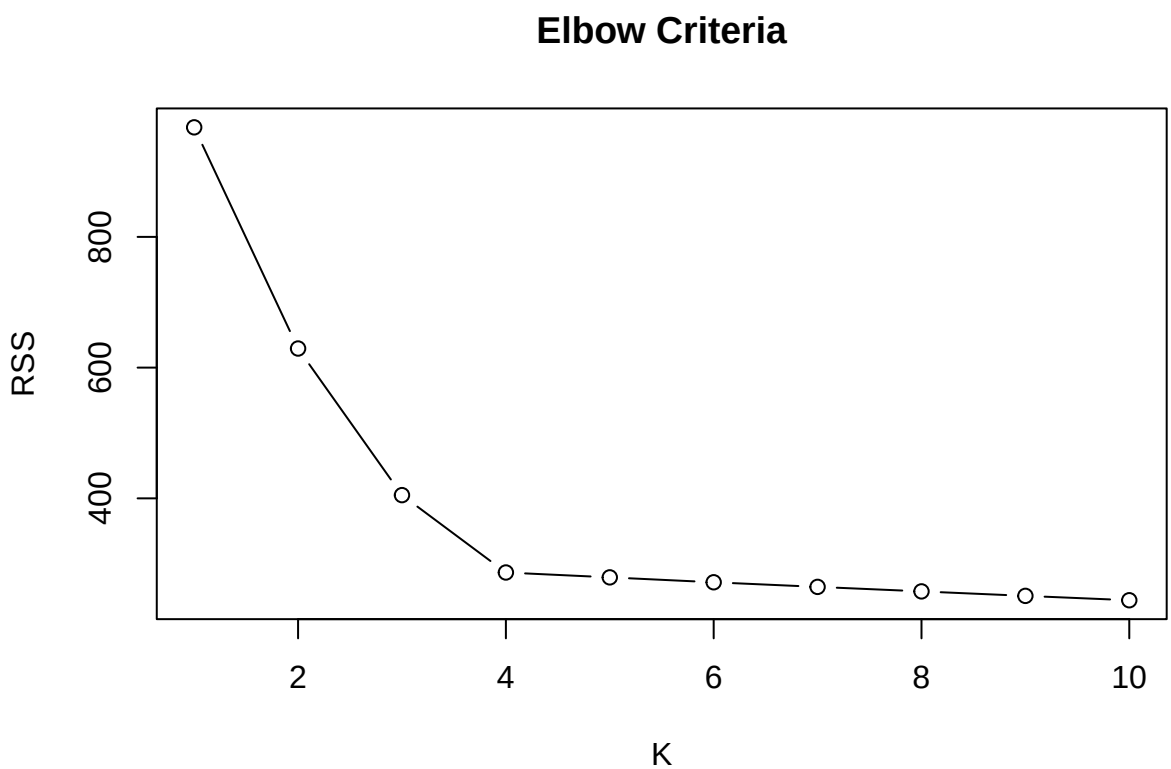
Choose the number of clusters

The algorithm can be run using only one k every time. Run k $(= 1, \ldots, K)$ times SSMF algorithm using `for` loop and save the results to a list, then 'elbow criteria' or the gap statistic is processed in the next step.

```
# Set the maximum k that is used in loop.
K <- 10

# Run the SSMF algorithm with various k and save the
# results as a list
fit_SSMF <- list()
for (k in 1:K) {
    fit <- ssmf(X, k = k, lr = 0.001, meth = "kmeans")
    fit_SSMF[[k]] <- fit
}
```

Plot the residual sum of squares ("elbow criteria")

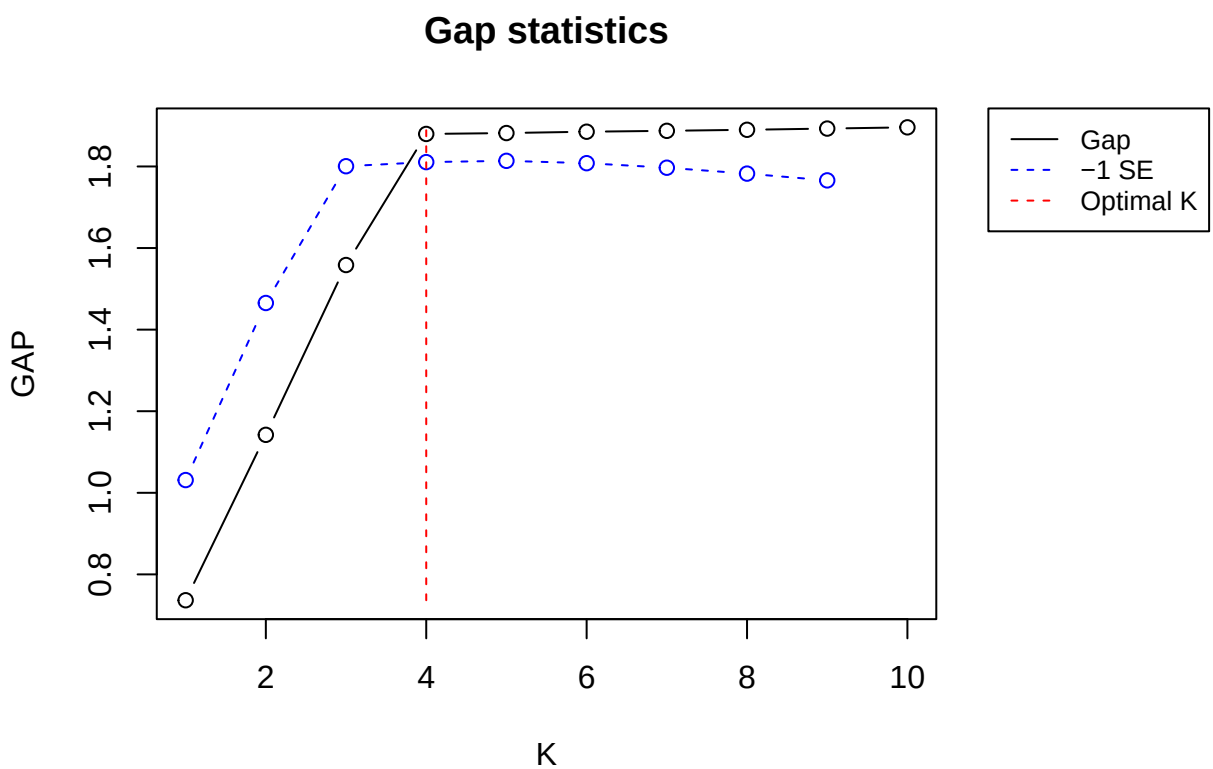
```
# Extract the RSS and save as a vector
rss_SSMF <- unlist(lapply(fit_SSMF, function(x) x$SSE))

# Plot RSS
plot(1:K, rss_SSMF, type = "b", xlab = "K", ylab = "RSS", main = "Elbow Criteria")
```



If the 'elbow' is not clear, the gap statistic is suggested.

Gap statistic

```
# Apply the gap statistics to the simulated data
fit_gap <- gap(X, rss = rss_SSMF)

# Visualise the results
par(mar = c(5.1, 4.1, 4.1, 8.1), xpd = TRUE)
plot(x = c(1:K), y = fit_gap$gap, type = "b", ylim = range((fit_gap$gap),
    (fit_gap$gap - fit_gap$standard.error)[-1]), xlab = "K",
    ylab = "GAP", main = "Gap statistics")
lines(x = c(1:(K - 1)), y = (fit_gap$gap - fit_gap$standard.error)[-1],
    col = "blue", lty = 2, type = "b")
lines(x = rep(fit_gap$optimal.k, 2), y = range(fit_gap$gap),
    lty = 2, col = "red")
legend("topright", inset = c(-0.35, 0), col = c("black", "blue",
    "red"), lty = c(1, 2, 2), legend = c("Gap", "-1 SE", "Optimal K"),
    cex = 0.8)
```



The estimated number of prototypes is 4 from both the 'elbow criteria' and gap statistic.

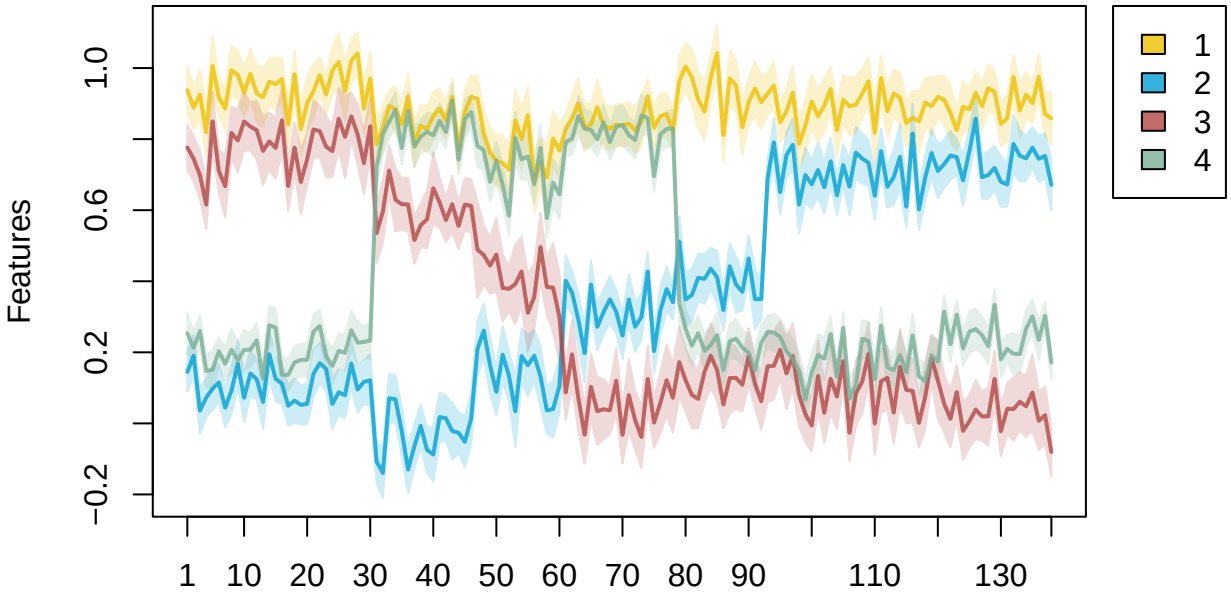
Bootstrap

```
# Set the number of times to bootstrap
M <- 1000

# Bootstrap Initial H (membership) matrix to start the
# algorithm Based on the results above, k=4 here
initialisation <- init(data = X, k = 4, method = "kmeans")
fit_boot <- bootstrap(data = X, k = 4, H = initialisation$H,
    mtimes = M)
```

Visualise the values of the bootstrap estimation and confidence intervals

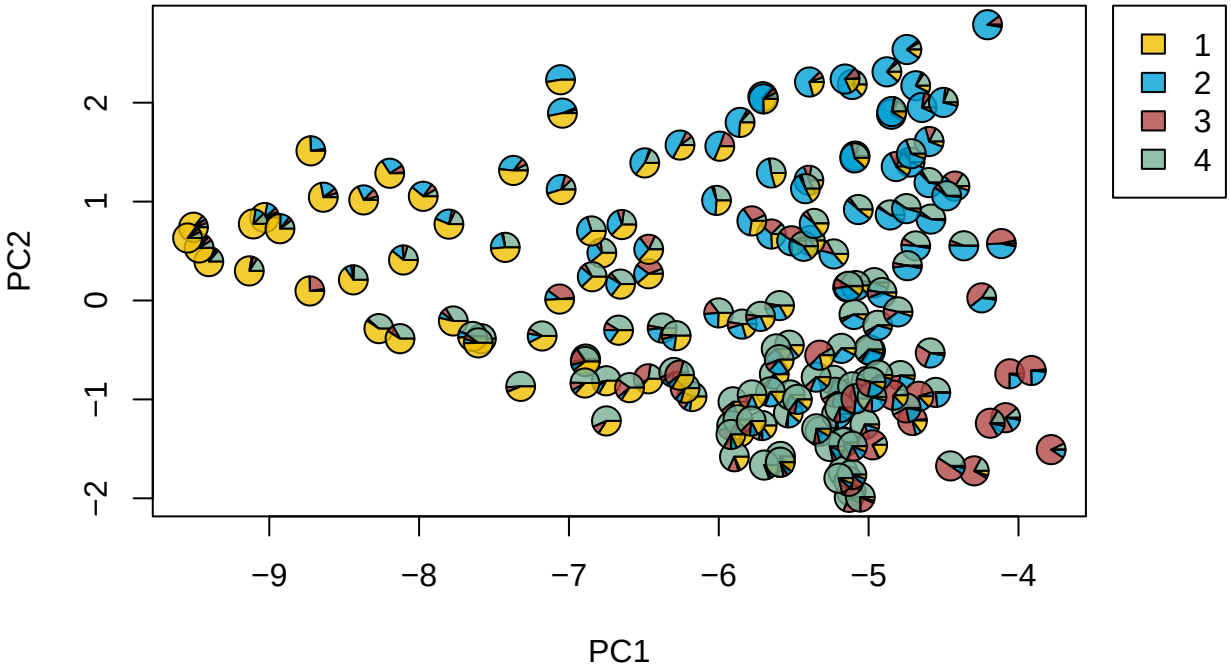
```
par(mar = c(5.1, 4.1, 4.1, 4.1), xpd = TRUE)
matplot(t(fit_boot$W.est[c(2, 4, 1, 3), ]), type = "n", xaxt = "n",
    ylab = "Features", ylim = range(fit_boot$lower, fit_boot$upper))
for (i in 1:4) {
    alpha <- 0.2
    plat <- c(pal_jco("default", alpha = alpha)(7)[2], pal_jama("default",
        alpha = alpha)(7)[c(3:5)])
    col <- switch(i, plat[1], plat[2], plat[3], plat[4])
    polygon(c(1:138, 138:1), c(fit_boot$upper[c(2, 4, 1, 3),
        ][i, ], rev(fit_boot$lower[c(2, 4, 1, 3), ][i, ])), col = col,
        border = col, lty = 1)
}
matplot(t(fit_boot$W.est[c(2, 4, 1, 3), ]), type = "l", lty = 1,
```

```
    ylab = "Features", add = T, col = color, lwd = 2)
legend("topright", inset = c(-0.15, 0), legend = 1:4, fill = color[1:4])
axis(1, at = c(1, seq(10, 130, 10), 138), labels = c(1, seq(10,
    130, 10), 138))
```



Visualise the values of the estimated soft membership matrix

```
par(mar = c(5.1, 4.1, 4.1, 4.1), xpd = TRUE)
pies(lapply(apply(fit_SSMF[[4]]$H, 1, list), unlist), color.table = nv(color[1:4],
    c(4, 3, 2, 1)), x0 = pca_X$x[, 1], y0 = pca_X$x[, 2], radii = 2,
    ylab = "PC2", xlab = "PC1")
legend("topright", inset = c(-0.15, 0), legend = 1:4, fill = color[1:4])
```



Soft Adjusted Rand Index (sARI)

```r
# Compare the true and estimated soft membership matrix
sARI(fit_SSMF[[4]]$H, H)
```

```
## [1] 0.3439103
```

```r
# Self comparison of the true soft membership matrix
sARI(H, H)
```

```
## [1] 0.3816684
```

Shannon diversity index

```r
E <- rep(NA, nrow(fit_SSMF[[4]]$H))
for (i in 1:nrow(fit_SSMF[[4]]$H)) {
    E[i] <- diversity(fit_SSMF[[4]]$H[i, ], two.power = T)
}
round(mean(E), 1)
```

```
## [1] 2.7
```

The estimated soft memberships of the simulated observations have an average values of $2^{\mathrm{E}(h_{i\cdot})} = 2.7$, indicating that most observations softly clustered by SSMF are fuzzy in 2 clusters.

Addition NOTE

The SSMF algorithm has randomness, this document presents one of the situation using `set.seed(12345)` at the beginning. The results will be various with different seeds that set in R. It is suggest to remove the seed and run the algorithm multiple times using the same values of k (the number of prototypes). Then, select a result with the lowest RSS, save it to the list and continue processing next steps.